EBU6475 Microprocessor Systems Design

End-of-Module (EoM) Project Brief
(Version 1)

By Jonathan Loo

Coursework Title: Fully Embedded 2D-LiDAR based Autonomous Mobile Robot (AMR)
Coursework Type: Group project
Marks contribution: 21% of total module marks

1. Project Overview

This EoM project serves as the capstone for the Microprocessor Systems Design module. You will conceive, design, implement, and operate a fully embedded 2D-LiDAR-based AMR that functions under realistic, real-time, and resource-constrained conditions. The project reflects professional embedded-robotics engineering design and is inspired by the industry-grade AMR illustrated in Figure 1.

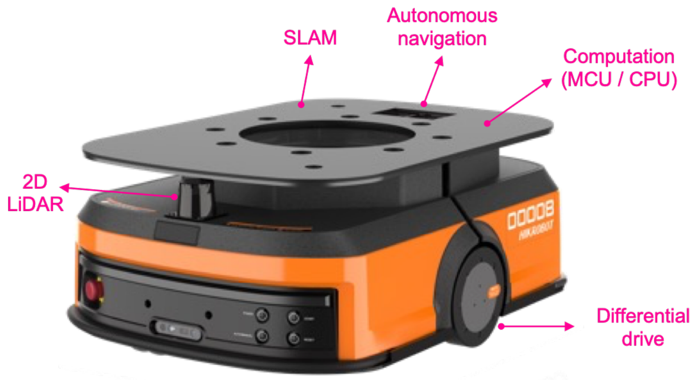

Figure 1. Industry-grade AMR

Your team will develop a complete **Sense → Think → Act** pipeline that executes entirely on the NUCLEO-F446RE (STM32F446RE microcontroller). All perception, localisation, mapping, planning, and control algorithms must run on-board; no external PC-based runtime computing is permitted.

Your AMR must demonstrate the following minimum behaviours:

- Autonomous exploration of an unknown maze
- 2D LiDAR–based environment perception
- Embedded localisation and mapping using odometry and scan data
- Goal-directed navigation from Start → Exit and Return-to-Start using the generated map
- Real-time closed-loop motor control under hardware and timing constraints

These capabilities must be implemented in a manner that reflects disciplined engineering judgement, hardware-software coherence, and real-time integration.

You will approach this project challenge using our GenAI–CDIO Authentic Assessment framework **(see Assessment Guidance on QMplus)**, which emphasises:

- The engineering process as the primary focus
- The final system product as evidence of that process
- Performance benchmarking as objective validation

The assessment evaluates your ability to:

- Interpret complex engineering challenges
- Develop feasible and realistic solutions within the project timeframe
- Design a coherent and technically sound embedded system architecture

EBU6475微处理器系统设计

模块结束（EoM）项目简报
（版本1）

作者：Jonathan Loo

课程名称：基于二维激光雷达的全嵌入式自主移动机器人（AMR）
课程类型：小组项目
评分贡献：占模块总分的21%

1. 项目概述

本工程硕士项目是微处理器系统设计模块的顶点课程。你将构思、设计、实现并操作一款完全嵌入式二维激光雷达自主移动机器人，使其在真实、实时且资源受限的条件下运行。该项目体现了专业嵌入式机器人工程设计，其灵感源自图1所示的工业级AMR。

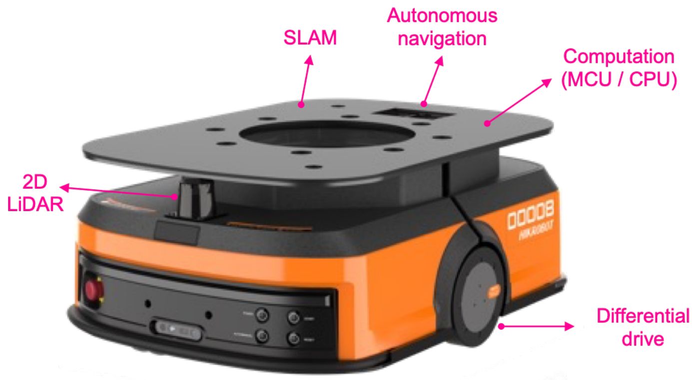

图1. 工业级抗菌材料

你的团队将开发一个完整的**感知→思考→行动**流程，该流程完全在NUCLEOF446RE（STM32F446RE微控制器）上执行。所有感知、定位、映射、规划和控制算法必须在板上运行；不允许使用基于PC的外部运行时计算。

您的AMR必须展示以下最低限度的行为：

- 未知迷宫的自主探索
- 基于二维激光雷达的环境感知
- 基于里程计与扫描数据的嵌入式定位与地图构建
- 基于生成地图的从起始点→出口及返回起始点的目标导向导航
- 硬件与时序约束下的实时闭环电机控制

这些功能的实现必须体现严谨的工程判断、软硬件一致性以及实时集成。

您将采用我们的GenAI– CDIO 真实评估框架（**参见QMplus评估指南）**来应对这一项目挑战，该框架强调：

- 以工程过程为研究重点
- 作为该过程证据的最终系统产品
- 将性能基准测试作为客观验证手段 该评估可评估您以

下能力：

- 解读复杂工程挑战
- 在项目时限内制定切实可行的解决方案
- 设计一个技术上合理且连贯的嵌入式系统体系结构

- Justify design trade-offs and limitations under practical constraints
- Validate performance using structured benchmarking
- Demonstrate engineering judgement through transparent, evidence-based reasoning

This project facilitates scaffolded and experiential learning. You will encounter intentional knowledge gaps designed to prompt investigation, critical thinking, and the exercise of engineering judgement. You are expected to apply structured reasoning across the full CDIO lifecycle: **Conceive → Design → Implement → Operate**.

You are encouraged to use GenAI tools to:

- Explore unfamiliar technical concepts (e.g., PID control, ICP scan matching, occupancy-grid mapping, A* search)
- Compare and evaluate alternative design approaches
- Support firmware development and debugging
- Reasoning about systems architecture and validation strategies

However, you remain fully responsible for:

- Verifying all AI-assisted outputs
- Ensuring technical accuracy and engineering validity
- Making and clearly justifying your own design decisions

This project is designed to develop your Engineering Competence (EC) through an experiential, end-to-end engineering workflow structured across the full CDIO lifecycle. Through systematic problem framing, design justification, implementation under constraints, and performance validation, you will progressively strengthen your ability to apply disciplined engineering reasoning in authentic contexts.

By the end of the module, you will have realised an integrated autonomous system, validated its behaviour through structured testing and benchmarking, and demonstrated your capacity to analyse, justify, and refine engineering decisions under realistic technical and resource constraints.

2. Standardised Hardware Kit

Each team receives a standardised hardware kit as shown in Figure 1 to ensure benchmarking fairness:

- STM32 NUCLEO-F446RE
- 2D LiDAR scanner (UART interface)
- MC520 Geared DC Motors with AT8236 motor driver (Timer/PWM interface)
- HC-04 Bluetooth module (UART interface)
- SSD1306 OLED display with 4 push buttons (I2C and GPIO interface)
- 3-gang Potentiometers (ADC interface)
- Logic analyser (signal/data analysis)
- 12V battery pack and charging adaptor

All teams will operate under identical hardware conditions to ensure fairness in quantitative benchmarking.

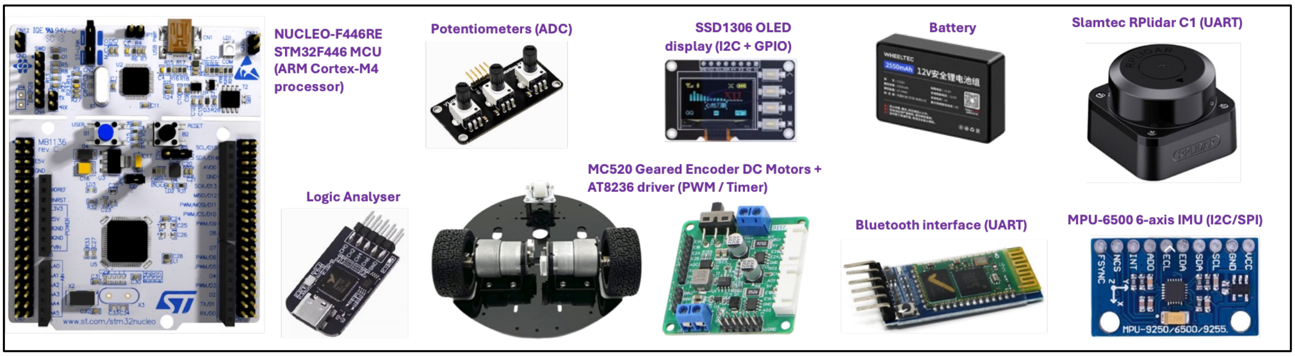

Figure 1. Standardised Hardware Kit

- 在实际约束条件下合理化设计权衡与局限性
- 通过结构化基准测试验证性能
- 通过透明、循证的推理展示工程判断力

本项目促进支架式和体验式学习。你将遇到有意设计的知识空白，旨在促进调查、批判性思维和工程判断的运用。你被期望在整个 CDIO 生命周期中应用结构化推理：**构思→设计→实施→操作**。

鼓励您使用GenAI工具：

- 探索不熟悉的技术概念（例如PID控制、ICP扫描匹配、占用网格映射、A*搜索）
- 比较和评估替代设计方法
- 支持固件开发与调试
- 关于系统架构与验证策略的讨论。然而，您仍需承担以下全

部责任：

- 验证所有AI辅助输出
- 确保技术准确性和工程有效性
- 制定并清晰论证自己的设计决策

本项目旨在通过贯穿 CDIO 全生命周期的体验式、端到端工程工作流程来提升你的工程能力（EC）。通过系统化的问题框架构建、设计论证、约束条件下的实施以及性能验证，你将逐步提升在真实场景中运用严谨工程推理的能力。

完成本模块后，您将构建出一套集成化的自主系统，通过结构化测试与基准测试验证其运行表现，并在实际技术与资源限制条件下，充分展现分析、论证及优化工程决策的能力。

2. 标准化硬件套件

为确保基准测试公平性，各团队均获得如图1所示的标准化硬件套件：

- STM32 NUCLEO-F446RE
- 二维激光雷达扫描仪（UART接口）
- MC520齿轮直流电机（配备AT8236电机驱动器，支持定时/ PWM 接口）
- HC-04蓝牙模块（UART接口）
- SSD1306 OLED 显示模块，配备4个按钮（支持I2C和 GPIO 接口）
- 三通道电位器（ADC接口）
- 逻辑分析仪（信号/数据分析）
- 12V电池组和充电适配器

所有团队将在相同的硬件条件下运行，以确保定量基准测试的公平性。

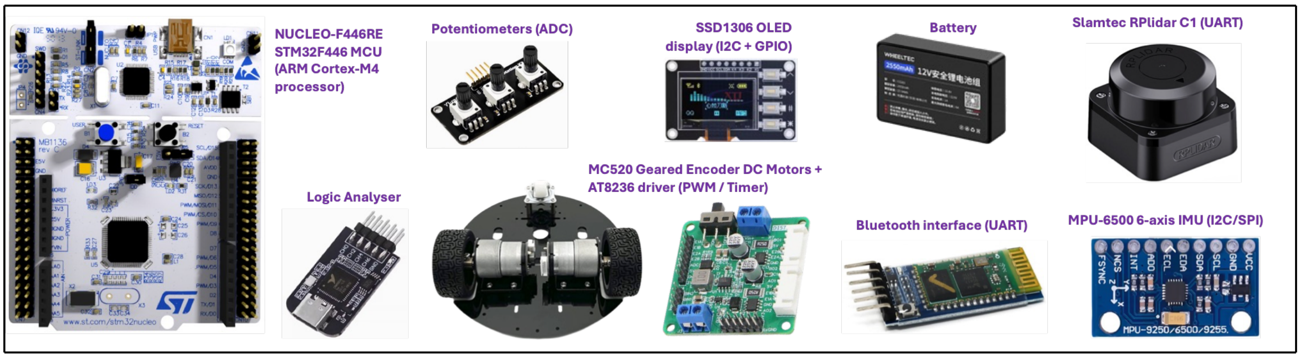

图1. 标准化硬件套件

3. Core Technical Requirements

Your system must implement a complete embedded autonomy pipeline structured around: **Sense → Think → Act**

All components must execute entirely on the STM32 microcontroller and operate under memory and real-time constraints.

Your system must satisfy the following minimum functional requirements.

a) Sense – Embedded Perception
- Acquire 2D LiDAR scan data via an appropriate UART interface
- Acquire wheel-encoder data using interrupt-based timer capture
- Implement basic filtering or pre-processing to manage sensor noise
- Sensing must be reliable, synchronised, and integrated into the real-time control pipeline

b) Think – Embedded Cognition
- Localisation and Mapping
  - Implement wheel-encoder-based odometry using differential-drive kinematics to estimate robot pose (x, y, θ) in real time
  - Implement a 2D Occupancy Grid Map (OGM) as the primary environment representation
  - Implement lightweight scan matching (e.g., ICP or equivalent) to refine pose estimation and mitigate cumulative odometry drift
  - Update the OGM incrementally using LiDAR data
  - Manage map resolution, memory allocation, and boundary conditions within embedded resource constraints
- Navigation and Planning
  - Implement autonomous exploration (e.g., frontier exploration) based on the global path planning (e.g., A* or equivalent) operating on the OGM representation
  - Implement local obstacle avoidance (e.g., DWA)
  - Demonstrate successful **Start → Exit** and **Exit → Start** navigation.

c) Act – Embedded Control
- Implement closed-loop motor control
- Implement PID-based velocity or position control
- Justify tuning methodology
- Maintain stability and responsiveness under real-time constraints.

d) User Interface – Embedded Interaction
- Implement an embedded user interface to support system configuration, monitoring, and safe operation.
- Use the OLED display to present key system states and real-time status information without blocking real-time tasks.
- Implement debounced button handling (interrupt- or polling-based) to support essential controls such as start, return-to-origin, emergency stop, and mode selection.
- Use ADC-based potentiometer input for adjustable parameters (e.g., control gains, speed limits), ensuring sampling does not interfere with sensing or control processes.
- implement Bluetooth-based telemetry for remote monitoring and debugging, ensuring that communication does not disrupt critical real-time execution.
- The user interface must support safe, intuitive interaction and provide meaningful visibility into system behaviour.

e) Real-Time Software Architecture
- Use interrupts and DMA for efficient, low-latency operation
- Use FreeRTOS with clearly separated tasks based on system function
- Real-time coordination across **Sense-Think-Act** must be explicit and measurable

3. **核心技术要求**

你的系统必须实现一个完整的嵌入式自主性流程，其结构围绕以下内容构建：**感知→思考→行动**

所有组件必须完全在STM32微控制器上运行，并在内存和实时性约束下操作。

您的系统必须满足以下最低功能要求。

a) 感觉——内嵌知觉
- 通过合适的UART接口获取二维激光雷达扫描数据
- 基于中断定时器捕获获取轮式编码器数据
- 实施基础滤波或预处理以管理传感器噪声
- 传感系统必须可靠、同步，并集成到实时控制流程中

b) 思考——嵌入式认知
- 定位与映射
  - 采用差速驱动运动学结合轮式编码器里程计技术，实时估算机器人位姿（x、y、θ）
  - 采用二维占用网格地图（OGM）作为主要环境表示形式
  - 采用轻量级扫描匹配技术（如ICP或等效方法）以优化姿态估计并缓解累积里程漂移
  - 利用激光雷达数据增量更新OGM
  - 在嵌入式资源约束条件下管理地图分辨率、内存分配及边界条件
- 导航与规划
  - 基于全局路径规划（如A*算法或等效算法）实现自主探索（例如前沿探索），该规划基于OGM表示进行操作
  - 实施局部障碍物规避（如DWA）
  - 演示成功的**启动 → 退出**和**退出 → 启动**导航。

c) 嵌入式控制
- 实现闭环电机控制
- 实现基于PID的速度或位置控制
- 合理化调校方法
- 在实时约束条件下保持稳定性和响应性。

d) 用户界面——嵌入式交互
- 实现嵌入式用户界面以支持系统配置、监控及安全运行。
- 通过 OLED 显示界面呈现关键系统状态和实时状态信息，同时不影响实时任务的执行。
- 采用去抖动按钮处理（基于中断或轮询）以支持启动、返回原点、紧急停止及模式选择等关键控制功能。
- 采用基于模数转换器（ADC）的电位器输入以实现参数可调（如控制增益、速度限制），确保采样不会干扰传感或控制过程。
- 采用基于蓝牙的遥测技术实现远程监控与调试，确保通信不会干扰关键实时执行。
- 用户界面必须支持安全、直观的交互操作，并提供对系统行为的有意义可见性。

e) 实时软件体系结构
- 采用中断和直接内存访问（DMA）实现高效低延迟操作
- 基于系统功能明确分离任务的FreeRTOS使用
- 跨**感知-思考-行动**的实时协调必须是明确的和可测量的

f) Safety and Fault Handling (few examples)

- Detect critical operational faults (e.g., LiDAR timeout, encoder stall, planner failure) and transition the system to a defined Fault state.
- Implement deterministic safe-stop behaviour under fault conditions, including controlled motor shutdown.
- Implement a latched hardware emergency stop that immediately disables PWM outputs and manual reset mechanism.
- Implement power integrity protection (e.g., low-voltage and brown-out detection) with graceful shutdown to prevent unstable behaviour.

g) Debugging, Validation and Data Visualisation

- Use structured debugging to verify sensing, mapping, planning, control, and task scheduling
- Use appropriate tools (SWV, logs, logic analyser, trace) to validate timing behaviour, interrupts, DMA, and FreeRTOS execution
- Validate performance under realistic conditions
- Record and visualise key parameters for tuning and benchmarking

4. Simulation Platform

Figure 2 presents a Python-based simulation platform is provided to support early-stage algorithm development and reasoning prior to hardware deployment. The simulator enables controlled design and testing of navigation strategies (e.g., exploration, global planning, and obstacle avoidance) within a maze environment, where start and exit positions may vary and the map is not pre-known.

The simulation environment supports conceptual understanding, structured experimentation, and validation of navigation logic before transferring solutions to the STM32 microcontroller platform.

Following simulation validation, students are required to re-implement and integrate the complete **Sense–Think–Act** pipeline as fully embedded firmware operating under real-time, memory, and processing constraints.

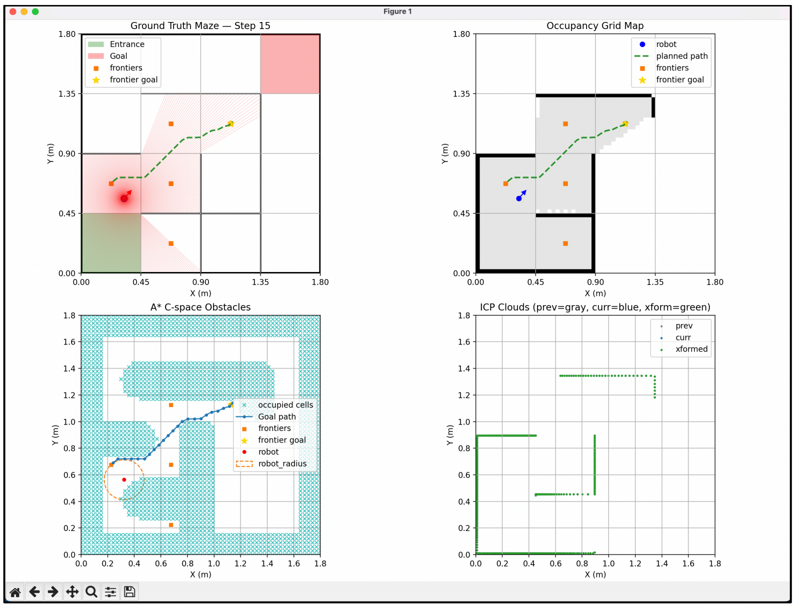

Figure 2. Python-based AMR simulation platform

This simulation-to-hardware workflow reflects professional engineering practice, where algorithms are first evaluated in controlled environments before being deployed on resource-constrained embedded systems. It supports coherent system development across algorithm design, firmware integration, and performance validation.

f) 安全与故障处理（示例）

- 检测关键操作故障（如激光雷达超时、编码器卡死、规划器失效）并将系统切换至预设故障状态。
- 在故障条件下实施确定性安全停机行为，包括受控电机停机。
- 实现带锁存功能的硬件紧急停止装置，可立即禁用 PWM 输出并提供手动复位功能。
- 实施电源完整性保护（如低压检测与褐光检测），并配备优雅关机功能以防止系统不稳定运行。

g) 调试、验证与数据可视化

- 采用结构化调试验证感知、映射、规划、控制及任务调度
- 使用相关工具（SWV、日志、逻辑分析器、跟踪工具）验证时序行为、中断、直接内存访问（DMA）及FreeRTOS的执行情况
- 在实际条件下验证性能
- 记录并可视化关键参数以进行调优与基准测试

4. 仿真平台

图2展示了一个基于Python的仿真平台，用于支持硬件部署前的早期算法开发与推理。该模拟器可在迷宫环境中对导航策略（如探索、全局规划和避障）进行受控设计与测试，其中起始与出口位置可能变化，且地图未预先确定。

该仿真环境支持概念理解、结构化实验及导航逻辑验证，随后将解决方案迁移至STM32微控制器平台。

在完成模拟验证后，学生需要重新实现并集成完整的**感知-思考-行动**流程，使其作为完全嵌入式固件运行，并在实时、内存和处理限制条件下工作。

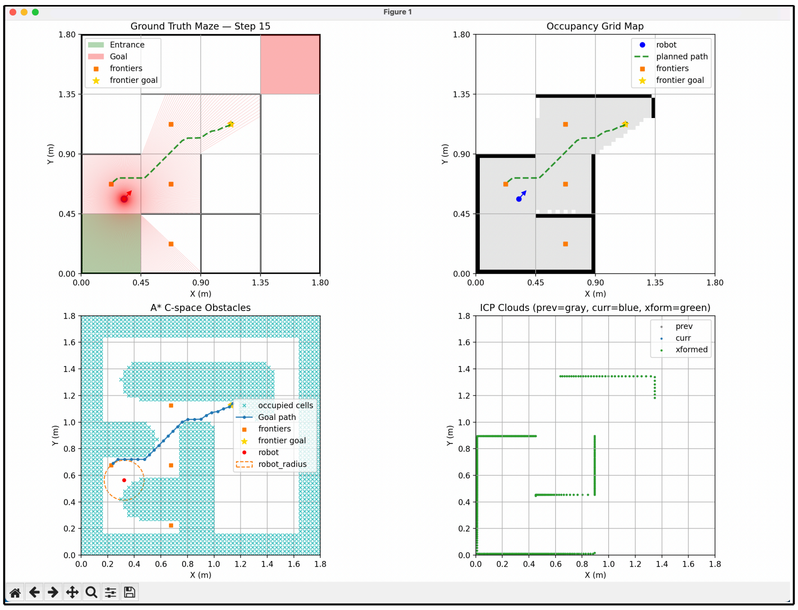

图2. 基于Python的抗菌药物耐药性（AMR）模拟平台

这种从仿真到硬件的工作流程体现了专业工程实践，即算法首先在受控环境中进行评估，然后部署到资源受限的嵌入式系统上。该流程支持算法设计、固件集成和性能验证之间的连贯系统开发。

5. Coursework Marking Criteria

This coursework is assessed using a Hybrid Qualitative-Quantitative model within the GenAI-CDIO Authentic Assessment framework.

The final coursework mark consists of:

- Qualitative (CDIO-based evaluation) – 80%
- Quantitative (Benchmark-based performance evaluation) – 20%

The final coursework mark (0-100) is calculated as:

$$Final = 0.8 \cdot S_{CDIO} + 0.2 \cdot S_{perf}$$

The detailed descriptions of each component are provided in Sections 5.1 and 5.2.

5.1. Qualitative (CDIO-based evaluation)

The qualitative component evaluates engineering competence across the structured CDIO lifecycle. Table 1 summarises the assessment criteria and the mark distribution for each stage within the framework.

Table 1

Marking Criteria	Description	Weight (%)
1. Conceive: Critical assessment of the project challenge	Assesses the ability to interpret project requirements, define objectives, establish appropriate specifications, and frame the engineering problem in a structured and defensible manner, considering practical constraints and real-world applicability.	20%
2. Design: Higher-order Thinking	Assesses the quality of design development, including critical analysis, evaluation of alternatives, trade-off justification, and coherence of the proposed system architecture. Emphasis is placed on defensible technical reasoning rather than feature complexity.	20%
3. Implement: System Realisation and Technical Integration	Assesses the translation of design into a functional system, including hardware-software integration, structured firmware development, coordination of subsystems, and technical execution consistent with stated design decisions.	20%
4. Operate: Verification, Validation, and Performance Evaluation	Assesses the ability to systematically test and validate system behaviour, analyse performance against defined requirements, apply appropriate debugging and monitoring techniques, and provide objective evidence of system functionality under realistic conditions.	20%
5. Reflection and Self-Assessment	Assesses critical reflection across all phases of the engineering process, including evaluation of design choices, validation outcomes, limitations, and responsible use of digital tools (e.g., GenAI) with evidence of verification, refinement, and learning progression.	10%
6. Teamwork and Collaborative Learning	Assesses effective collaboration, technical communication, responsible task ownership, integration of contributions, and shared accountability in delivering a coherent engineering solution.	10%

Note that:

- Criteria 1–4 are assessed via your documentation and video presentation. These criteria measure your project outcomes in terms of technical proficiency and practical knowledge applied to real-world challenges.

5. 课程作业评分标准

本课程作业采用GenAI- CDIO 真实评估框架中的混合定性-定量评估模型进行评分。

最终课程成绩由以下部分组成:

- 定性（基于 CDIO 的评估）– 80%
- 定量（基于基准的绩效评估）——20%

最终课程成绩（0-100分）计算公式为:

$$最终值 = 0.8 . S_{CDIO}+0.2.Sperf$$

各组件的详细描述见第5.1和5.2节。

5.1. 定性（基于 CDIO 的评估）

定性评估部分用于评价工程能力在整个结构化 CDIO 生命周期中的表现。表1汇总了该框架下各阶段的评估标准及评分分布。

表1

标记标准	描述	重量（%）
1. 设想： 项目挑战的批判性评估	评估项目需求解读能力、目标定义能力、规范制定能力，以及以结构化且可论证的方式构建工程问题的能力，同时兼顾实际约束条件与现实应用可行性。	20%
2. 设计：高阶思维	评估设计开发的质量，包括关键分析、备选方案评估、权衡论证以及所提议系统架构的连贯性。重点在于可辩护的技术推理而非功能复杂性。	20%
3. 实现：系统实现与技术集成	评估设计向功能系统的转化，包括硬件-软件集成、结构化固件开发、子系统协调以及与既定设计决策一致的技术执行。	20%
4. 操作： 验证、确认与性能评估	评估系统行为的系统性测试与验证能力，分析性能是否符合既定需求，应用适当的调试与监控技术，并在实际工况下提供系统功能的客观证据。	20%
5. 反思与自我评估	评估工程流程各阶段的关键性反思，包括对设计选择的评估、验证结果、局限性分析，以及数字工具（如GenAI）的负责任使用，并提供验证、优化和学习进展的证据。	10%
6. 团队合作与协作学习	评估有效协作、技术沟通、任务责任归属、贡献整合及共同问责制在提供连贯工程解决方案中的表现。	10%

请注意:

- 标准1-4通过您的文档和视频演示进行评估。这些标准从技术熟练度和实际应用知识应用于现实挑战的角度衡量项目成果。

- Criteria 5-6 are assessed during lab session where you will demonstrate your project process and progress, reflective activities, continuous feedback, and project quality improvement.

5.2. Quantitative (Benchmark-based performance evaluation)

The quantitative component assesses the objective performance of your system product (AMR) under standardised test conditions.

All teams will be evaluated using the maze layout shown in Figure 2, ensuring fairness, consistency, and transparency across all submissions.

The maze consists of a 5×5 grid, with each cell measuring 70 cm × 70 cm. The Start and Exit coordinates are predefined; however, the map itself is unknown and must be autonomously explored by your AMR.

Figure 2. Maze layout for AMR benchmarking

Your AMR's performance will be measured using two time-navigation metrics recorded while operating within the maze:

- Exit Time (Start → Exit) maximum 120s
- Return Time (Exit → Start) maximum 120s

Each metric is converted into a normalised performance score using clamped linear scaling:

$$P_i = \mathrm{clamp}\left(100 \cdot \frac{T_{Zero,i} - T_{Team,i}}{T_{Zero,i} - T_{Full,i}}, 0, 100\right)$$

where

- $T_{\mathrm{full},i}$: Full-credit time is $\leq$ 30s
- $T_{\mathrm{zero},i}$: Zero-credit time is $\geq$120s
- $T_{\mathrm{Team},i}$: Team's measured performance time

The final quantitative performance score is then computed as:

$$S_{perf} = 0.5 \cdot (P_{Exit} + P_{Return})$$

For example, with $T_{Team,Exit} = 80s$, $T_{Team,Return} = 60s$, the normalised scores are:

$$P_{Exit} = 100 \cdot \frac{120 - 90}{120 - 30} = 33$$

$$P_{Return} = 100 \cdot \frac{120 - 60}{120 - 30} = 67$$

$$S_{perf} = 0.5 \cdot (P_{Exit} + P_{Return}) = 50$$

- 标准5-6将在实验室环节中进行评估，届时您需展示项目流程与进展、反思性活动、持续反馈及项目质量改进情况。

5.2. 定量（基于基准的绩效评估）

定量评估部分用于测定您的系统产品（AMR）在标准化测试条件下的客观性能表现。

所有团队将采用图2所示的迷宫布局进行评估，以确保所有提交结果的公平性、一致性和透明度。

该迷宫由一个5×5网格组成，每个单元格尺寸为70厘米×70厘米。起始点和出口坐标已预先设定，但地图本身未知，需由您的自主移动机器人（AMR）自主探索。

图2. AMR基准测试的迷宫布局

将通过在迷宫内运行期间记录的两项时间导航指标来评估您的AMR性能：

- 退出时间（开始→退出）最长120秒
- 返回时间（退出→开始）最长120秒

每个指标均采用钳制线性缩放法转换为标准化绩效评分：

$$P_i\text{=clamp}\left(100\cdot\frac{T_{Zero,i} - T_{Team,i}}{T_{Zero,i} - T_{Full,i}},\ 0,\ 100\right)$$

其中

- $T_{全}$时，i：全学分时间为≤ 30秒
- $T_{零}$，i：零信用时间为≥120秒
- $T_{团队}$，i：团队的测量性能时间 最终的定量绩效分数计算

公式为：

$$S_{perf} = 0.5 \cdot (P_{Exit} + P_{Return})$$

例如，对于T_{Team}，Exit=80秒，T_{Team}，$_{Return}$=60秒，标准化分数为：

$$P_{Exit} = 100 \cdot \frac{120 - 90}{120 - 30} = 33$$

$$P_{Return} = 100 \cdot \frac{120 - 60}{120 - 30} = 67$$

$$S_{perf} = 0.5 \cdot (P_{Exit} + P_{Return}) = 50$$

6. Project deliverables

Your team is required to submit the following deliverables as evidence of your engineering process, system realisation, and validated performance.

6.1. Technical Documentation Requirements (PDF, Maximum 10 Pages)

Your Technical Documentation should present clear, well-structured evidence showing how your work satisfies the marking criteria for each phase of the CDIO lifecycle (Conceive, Design, Implement, Operate) as defined in Table 1.

For each CDIO phase, your documentation must clearly address the following requirements:

- Explain the lifecycle progression by describing how the work completed in this phase builds on the previous phase and how it informs the next phase within the overall project development process.
- Provide concrete supporting evidence by including material such as problem formulation, system-architecture justification, implementation excerpts, validation data, or measured performance results that directly relate to this phase.
- Justify your engineering decisions by clearly explaining the key choices made during this phase and detailing the methods used to test, validate, or refine those decisions.
- Demonstrate the use of GenAI tools by presenting:
    - One example of a useful GenAI output that improved your reasoning
    - One example of an unhelpful or misleading output that you verified, corrected, or refined during your work

6.2. Video Submission Requirements

Two separate videos must be submitted as part of this coursework:

- Product Quality Video (Maximum 7 Minutes)
- Product Performance Benchmark Video (Separate Submission)

Each video serves a distinct assessment purpose within the hybrid qualitative–quantitative framework.

6.2.1. Product Quality Video (Maximum 7 Minutes)

This video forms part of the **Product Quality (Qualitative) Assessment** under the CDIO-based evaluation. It assesses implemented system features, integration quality, engineering reasoning, validation discipline, and real-time architecture.

This video evaluates the quality and coherence of the implemented system. It does not assess benchmark timing performance.

The video must be structured as follows:

a) Team Introduction and Responsibilities (Approx. 1 Minute)
    - Introduce all team members
    - Briefly describe individual responsibilities and technical contributions
    - Explain how responsibilities were coordinated across the overall system architecture

b) System Product Demonstration Approx. 2 Minutes): Demonstrate the implemented technical requirements under realistic operating conditions. This must clearly show:
    - Functional implementation of required features
    - Coherent integration across sensing, cognition, control, and user interface components.
    - Observable cause-effect behaviour (e.g., sensor input or user interaction leading to predictable system response)
    - Stable and reliable autonomous operation
    - This section evaluates system integration and functional completeness, not timing performance or detailed validation evidence

6. 项目交付物

贵团队需提交以下交付成果，用以证明工程流程、系统实现及经验证的性能。

6.1. 技术文档要求（PDF格式，最多10页）

贵方技术文档需提供清晰、结构严谨的证据，证明贵方工作符合表1所定义的 CDIO 生命周期各阶段（构思、设计、实施、运营）的评分标准。

对于每个 CDIO 阶段，您的文档必须明确说明以下要求：

- 通过描述本阶段完成的工作如何在整体项目开发过程中建立在前一阶段的基础上，并为下一阶段提供信息，来解释生命周期进展。
- 提供具体的支持性证据，包括与本阶段直接相关的材料，如问题表述、系统架构论证、实施摘录、验证数据或实测性能结果。
- 通过清晰解释此阶段的关键决策，并详细说明用于测试、验证或优化这些决策的方法，来证明你的工程决策的合理性。
- 通过展示：
  - 一个实用的GenAI输出示例，可提升你的推理能力
  - 在工作中核实、修正或优化的无效或误导性输出示例

6.2. 视频提交要求

本课程作业需提交两段独立视频：

- 产品质量视频（最长7分钟）
- 产品性能基准视频（单独提交）

在混合定性-定量框架中，每个视频均具有独特的评估目的。

6.2.1. 产品质量视频（最长7分钟）

本视频是基于 CDIO 的评估体系中**产品质量（定性）评估**的组成部分，主要评估已实施的系统功能、集成质量、工程论证、验证规范及实时架构。

本视频评估了所实施系统的质量与一致性，但未对基准时间性能进行评估。

视频必须按照以下结构组织：

a) 团队介绍与职责（约1分钟）
   - 介绍所有团队成员
   - 简要说明个人职责与技术贡献
   - 阐述系统架构中职责协调机制的运作方式

b) 系统产品演示（约2分钟）：在实际运行条件下展示已实现的技术要求。必须清晰展示：
   - 功能实现所需功能
   - 跨感知、认知、控制及用户界面组件的协同整合。
   - 可观测因果行为（例如，传感器输入或用户交互导致可预测的系统响应）
   - 稳定可靠的自主运行
   - 本节评估系统集成度与功能完整性，而非时间性能或详细验证证据

c) Product Evaluation (Approx. 2 Minutes): Demonstrate how system behaviour was verified, validated and refined through structured engineering practice. This may include:
   - Monitoring outputs or logged parameters
   - Real-time system state visualisation
   - Evidence of testing, refinement, and optimisation
   - This section evaluates validation discipline and evidence-based engineering reasoning

d) Code (Approx. 2 Minutes): Explain the overall firmware structure and system organisation. Demonstrate:
   - Task organisation and scheduling strategy
   - Use of interrupts, DMA, and FreeRTOS.
   - Coordination across the Sense-Think-Act pipeline.
   - Key architectural decisions supporting real-time execution.
   - The emphasis should be on system-level clarity rather than detailed code walkthrough.

6.2.2. Product Performance Benchmark Video

This video forms part of the **Product Performance (Quantitative) Assessment (20%)** and is used solely to determine benchmark-based performance scores.

The video must:

- Record the complete Start → Exit run
- Record the complete Exit → Start return run
- Display a visible and readable timer throughout the recording

Each navigation phase is assessed independently under the 120s maximum limit (see Section 5.2 for the marking calculation).

The video may be shown at high speed (e.g., 4× or 8×), provided the timer remains clearly visible throughout. The recording must be continuous, with no cuts or edits.

The benchmark video must provide clear and traceable evidence of performance under standardised testing conditions.

6.2.3. Complete Project Code (in Zip)

You must submit a single ZIP file containing all source code, header files, configuration files, and any supporting resources required to compile and run your system on the NUCLEO-F446RE.

This submission must correspond exactly to the system demonstrated in your Product Quality Video.

You must include a **short README** file within the ZIP. The README should briefly describe:

- The overall folder structure of your project
- Any custom modules or hardware abstraction components implemented

This submission serves as supporting evidence for the functionality and behaviour demonstrated in your submitted videos.

7. Importance Dates

- Project Brief Release: **Monday (Week 3)**
- Project Start: **Monday (Week 4)**
- Project progress check: **Friday (Week 7)**
- Project live demo and interview: **TBA (Week 13)**
- Project Submission to QMPlus: **Monday (Week 14)**

-------- End of Document --------

c) 产品评估（约2分钟）：展示系统行为如何通过结构化工程实践进行验证、确认和优化。内容可能包括：
   - 监测输出或记录参数
   - 实时系统状态可视化
   - 测试、改进和优化的证据
   - 本节评估验证规范与循证工程推理

d) 代码（约2分钟）：阐述固件整体架构与系统组织结构。
   演示：
   - 任务组织与调度策略
   - 中断、DMA与FreeRTOS的应用
   - 感知-思考-行动流程的协调
   - 支持实时执行的关键架构设计。
   - 重点应放在系统层面的清晰性，而非详细的代码走查。

6.2.2. 产品性能基准视频

该视频是**产品性能（定量）评估（20%）**的一部分，仅用于确定基于基准的性能分数。

视频必须：

- 完整记录启动→退出运行过程
- 记录完整的退出→启动返回运行
- 在整个记录过程中显示可见且可读的计时器

每个导航阶段均在120秒最大时限内独立评估（标记计算方法参见第5.2节）。

该视频可以高速播放（例如4倍或8倍速），前提是计时器全程保持清晰可见。录像必须连续进行，不得剪辑或编辑。

基准视频必须在标准化测试条件下提供清晰且可追溯的性能证据。

6.2.3. 完整项目代码（压缩包）

您需提交一个ZIP压缩包，其中包含所有源代码、头文件、配置文件以及在 NUCLEO -F446RE上编译和运行系统所需的全部支持资源。

该提交必须与您的产品质量视频中展示的系统完全一致。您必须在ZIP中包含一个**简短的readme**文件。

readme应简要描述：

- 项目文件夹的整体结构
- 任何已实现的自定义模块或硬件抽象组件

本提交文件作为支持证据，用于证明您提交视频中展示的功能性与行为表现。

7. 重要日期

- 项目简报发布日期：**周一（第3周）**
- 项目开始日期：**周一（第4周）**
- 项目进度检查：**星期五（第7周）**
- 项目现场演示和面试：**待定（第13周）**
- 项目提交至QMPlus：**周一（第14周）**

-------- 文档结束 --------